Multi-sensor dataset of a tendon-driven continuum robot in dynamic motion

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Abstract

TO CHANGE This data descriptor presents a multi-sensor dataset on the quasi-static and dynamic motion of a tendon-driven continuum robot, including scenarios with and without contact with objects in the workspace. The robot is instrumented with a motion capture system, fiber Bragg grating sensors, cable tension gauges, motor encoders, a base force-torque sensor, inertial measurement units, and a contact force-torque sensor. The dataset covers planar, circular, and Lissajous trajectories at multiple speeds, organized in four batches: quasi-static motions, dynamic motions, dynamic motions with body-mounted inertial measurement units, and dynamic motions with contact. Both raw sensor recordings and processed data (filtered, time-aligned, uniformly resampled) are provided. The dataset enables benchmarking of dynamics models, training of data-driven methods, study of contact interactions, and sensor fusion research for continuum robots.

1 Background & Summary

Continuum robots are manipulators whose body undergoes continuous elastic deformation, actuated by tendons, pneumatics, or other distributed mechanisms [1, 2]. Their dynamics have been studied through various modeling approaches [3, 4, 5, 6], and there is growing interest in applying machine learning methods, in both statics [7] and dynamics [8], as well as studying their contact and interaction with the environment [9] with new studies addressing the contact kinematics with differentiable properties [10]. However, experimental validation in continuum robotics typically relies on ad-hoc prototypes instrumented with lab-specific sensors, and the resulting data is rarely made publicly available. A related effort produced a dataset for concentric tube robots [11], but no publicly available dataset currently combines kinematics and force measurements for a tendon-driven continuum robot undergoing both quasi-static and dynamic motions, including contact with objects in the workspace. This limited availability of open datasets prevents reproducibility of results and the comparison of competing modeling, control, and learning approaches on common ground.

This data descriptor presents a multi-sensor dataset recorded on a single-segment tendon-driven continuum robot. Kinematic measurements are provided by an OptiTrack motion capture system (disk poses), FBGS (distributed strain and shape), motor encoders (tendon displacements), and IMUs (angular velocities and linear accelerations). Force measurements are provided by Mark10 gauges (cable tensions), an ATI force-torque sensor (base wrench), and a Resense force-torque sensor (contact forces). The dataset is organized in four subsets: (i) quasi-static motions, (ii) dynamic motions, (iii) dynamic motions with IMUs mounted on the robot body, and (iv) dynamic motions with contact with an object in the workspace. Trajectories for dynamic motions include planar sweeps along x, y, and xy, circular paths, and Lissajous figures, each executed with two maximum angular velocities for the actuators: slow (1 $\mathrm{rad\,s^{-1}}$) and fast (3 $\mathrm{rad\,s^{-1}}$). Both raw sensor recordings and processed (filtered, resampled, time-aligned) data are provided.

The dataset is intended for benchmarking dynamics models against experimental data, training and evaluating data-driven methods, studying contact and interaction forces, investigating sensor fusion approaches, and performing parameter identification. Because the robot is built from commercially available components and the design is fully documented, the dataset can also serve as a reference for groups building similar platforms.

In total, the dataset comprises N experiments totaling approximately X hours of multi-sensor recording (Y GB of raw data, Z GB processed).

2 Methods

In this section, we define the methodology used to create the presented dataset, starting from the robot design and components to the specifics of the sensors. Finally, we describe how the data is collected giving a general idea of the data collection pipeline, the programming language and the use of CPU wall-time timestamps.

2.1 Design

2.2 Sensors

Different sensors have been mounted on the robot to capture different aspect of the motion, being it kinematics like position, angular velocities and linear accelerations, shape of the robot and pulled lenght of the cables, or force measures like tendon tensions, forces and torques at the base and at the contact with an object in the environment. In this section, we describe the characteristics of each sensor in terms of frequency readings, range of outputs, and a summary of their specifics available on their datasheet.

2.2.1 Mark 10

Cable tension was measured using Mark-10 M3-5 digital force gauges (Mark-10 Corporation, Copiague, NY, USA; `https://mrmmetrology.com/product/m3-5-mark-10-digital-force-gauge/`), with a full-scale capacity of 5 lbf ($\approx$ 22.2 N), a resolution of 0.005 lbf ($\approx$ 0.02 N), and a manufacturer-specified accuracy of $\pm$ 0.3% of full scale. All units were calibrated by an ISO 17025 accredited laboratory with an expanded uncertainty of $\pm$ 0.007 lbf (k=2, 95% coverage). Each gauge was interfaced to the acquisition computer via a serial connection at 115 200 baud and polled continuously using a request–response protocol. The effective sampling rate, limited by the serial round-trip latency, was approximately 335 Hz per gauge. The output was recorded directly in newtons.

2.2.2 FBGS

Distributed strain sensing along the robot backbone was performed using an FBGS multicore fiber (MCF) shape sensing system (FBGS International, Geel, Belgium; `https://www.fbgs.com`). The sensing fiber (ID: F13421-0007-003) is a four-core MCF with an outer diameter of 250 µm, featuring draw-tower gratings (DTGs) inscribed during the fiber drawing process. Each core contains 26 FBG gratings uniformly spaced at 20 mm intervals, yielding an active sensing length of 500 mm. The individual grating wavelengths span approximately 1512 nm to 1587 nm within the C-band, with a minimum inter-grating spacing of 0.8 nm. The core-to-center distance is 36.9 µm and the gauge factor is $0.777\mu\varepsilon^{-1}$. Three of the four cores are used for shape reconstruction (core 2 omitted), enabling the decoupling of bending curvature, bending direction, and temperature along the fiber length.

The fiber was interrogated by an FBG-Scan 904 spectrometer-based interrogator (FBGS), which features four optical channels over a 1510 nm to 1590 nm wavelength range with LC/APC connectors. The interrogator provides a wavelength precision of 0.3 pm (1σ), a wavelength linearity of 3 pm, an absolute wavelength accuracy of 10 pm, and a dynamic range exceeding 30 dB. Raw spectral data were processed by the ILLumiSense Shape CORE software (v1.1, FBGS), which runs as a background process on the acquisition computer, receives spectra from the interrogator via USB, performs peak detection, and computes curvature, bending angle, three-dimensional shape, and temperature at each grating location. The reconstructed shape is output as three-dimensional Cartesian coordinates with 1 mm arc-length resolution. With shape reconstruction enabled, the maximum achievable output rate of the Shape CORE software is 100 Hz, which was used throughout all experiments.

Shape CORE streams its output over a TCP connection on the local machine (port 50012). A custom C++ client connects to this stream, receives curvature, bending angle, and shape arrays at each time step, timestamps each sample using the system high-resolution clock (`std::chrono::high_resolution_clock`), and stores all data in memory. At the end of the recording, the buffered data is written to a CSV file. In this work, we only account for the reconstructed shape, while the curvature and bending angle can be used as detailed in [12].

Prior to measurements, the fiber was calibrated following the manufacturer's two-step procedure: a straight calibration to establish the reference wavelengths in the absence of bending strain, and a planar calibration to determine the reference twist of the fiber.

The manufacturer specifies the position of the last grating with a tolerance of ± 5 mm. To determine this position with millimetre-level accuracy, a hot-wire contact was applied along the fibre starting from the distal end and advanced towards the base. Because the interrogator reports strain and temperature at each grating location, the last grating that registered a change in temperature upon contact was identified as the distal sensing point. This procedure yielded the arc-length coordinate of the last measured grating to within ~ 1 mm, providing an accurate reference for shape reconstruction along the robot backbone. Once the distal grating was located, it was aligned with the tip disk of the continuum robot. The fibre was then fixed in place at both the base and the tip disk, ensuring a stable and repeatable routing along the backbone for all subsequent experiments.

2.2.3 ATI-FT

External force and torque measurements at the robot base were acquired using an ATI Mini40 six-axis force/torque sensor (calibration SI-40-2; ATI Industrial Automation, Apex, NC, USA; `https://www.ati-ia.com/products/ft/ft_models.aspx?id=mini40`). The Mini40 is a monolithic transducer machined from hardened stainless steel via electrical discharge machining (EDM), instrumented with silicon strain gages that provide a signal-to-noise ratio approximately 75 times higher than conventional foil gages. The transducer has a diameter of 40 mm, a height of 12.2 mm, and a mass of 49.9 g. With the SI-40-2 calibration, the sensing ranges are ± 40 N in F_x and F_y, ± 120 N in F_z, and ± 2 N m in T_x, T_y, and T_z. Single-axis overload ratings are ± 810 N (F_{xy}) and ± 2400 N (F_z), and the resonant frequencies are 3200 Hz (F_x, F_y, T_z) and 4900 Hz (F_z, T_x, T_y). The transducer was connected in the DAQ F/T configuration, with the six analog voltage channels (± 10 V range) digitized by a National Instruments data acquisition card using the `nidaqmx` Python driver. The hardware sampling clock was set to 1000 Hz; every five consecutive samples per channel were averaged to yield an effective wrench output rate of approximately 200 Hz. A 6×6 calibration matrix provided by ATI was applied in software to convert the bias-subtracted voltage readings into calibrated force and torque values (F_x, F_y, F_z, T_x, T_y, T_z).

2.2.4 Gyroscopes

TODO

2.2.5 Actuators

The tendons are actuated by four brushless DC motors (Cybergear, Xiaomi, China) with custom 3D-printed tendon spools of winding radius 20 mm, i.e., the radius at which the tendon cable is coiled around the spool. Each motor is a quasi-direct-drive (QDD) module integrating a three-phase, 28-pole frameless torque motor with a 7.75:1 planetary gearbox, a 14-bit single-turn absolute encoder, and onboard FOC electronics, in a compact package of 80.5 mm diameter and 36.5 mm height (317 g). The motors are powered at 24 V DC and provide a continuous output torque of 4 N m and a peak torque of 12 N m. Communication with the host PC is handled via a CAN 2.0 bus at 1 Mbit s^{-1}. The four motors are operated as two antagonistic pairs (motors 1&3 and 2&4), each pair controlling one planar degree of freedom of the robot tip. They are operated in a closed-loop position control scheme, where target shaft angles θ^* are commanded directly over CAN, with shaft angle, velocity, and output torque returned as real-time feedback at each control cycle. Angles are computed relative to the home position, which is initialized as the reading angle at the motor startup. The resulting tendon displacement $\Delta l = r_s \theta^*$, where $r_s = 20$ mm is the spool winding radius, is a geometric byproduct of the motor rotation.

2.2.6 Motion Capture System

The shape of the robot backbone and the pose of objects in contact with it are measured using an optical motion capture system (OptiTrack, NaturalPoint, USA) composed of five PrimeX13 and two PrimeX13W cameras. The PrimeX13 cameras feature a 1.3 MP sensor (1280×1024 px), a native frame rate of 240 Hz, a field of view of $56^\circ \times 46^\circ$, and a 3D positional accuracy of ± 0.15 mm. The two PrimeX13W cameras share the same sensor and frame rate but provide a wider field of view of $82^\circ \times 70^\circ$, suited to ensure full coverage of the compact workspace, at a slightly reduced accuracy of ± 0.30 mm. All cameras use 850 nm infrared illumination and communicate via GigE/PoE. Five rigid bodies composed of passive

retroreflective markers (≈6 mm diameter) are attached to equally-spaced disks along the robot backbone at arc-length positions $s \in \{0, 12, 24, 36, 48\}$ cm, providing discrete samples of the backbone pose. An additional rigid body is mounted on the contact object to track its 6-DoF pose within the workspace. The system runs on a dedicated host PC managed by Motive 3.4, which processes and solves 3D pose at 120 Hz. Data are streamed in real time to the dataset acquisition PC via a point-to-point Ethernet link using the NatNet 4.4 SDK in unicast UDP mode. A custom Python client receives each frame and logs the 6-DoF pose — position (x, y, z) and orientation (q_x, q_y, q_z, q_w) — of each rigid body, together with a per-body tracking validity flag. Timestamps are assigned on the acquisition PC using the system wall clock (`time.time()`), rather than Motive's internal camera exposure timestamps, which could induce a small and variable network transmission jitter in the logged timing.

2.2.7 Resense FT

Contact forces between the robot and objects in the workspace were measured using a Resense HEX 12 six-axis force/torque sensor (ordering code HEX125S-020S-1R1-1; Resense GmbH, Klingenberg, Germany; `https://www.resense.io`). The HEX 12 is a compact transducer machined from Titanium Grade 5, instrumented with foil strain gauges, with a diameter of 12.5 mm, a height of 11.0 mm, and a mass of 3.3 g. The nominal measurement ranges are ±25 N in F_x, F_y, and F_z, and ±125 mN m in T_x, T_y, and T_z, with a manufacturer-specified accuracy of 1% of full scale, a crosstalk of 3%, and an overload capacity of 300%. The sensor was paired with the Resense miniature evaluation electronics board (EVAL 100S-06-1, $54 \times 25 \times 12.6$ mm), which digitises the strain gauge signals with a 24-bit ADC, applies the manufacturer-provided calibration matrix onboard, and outputs calibrated forces (in newtons) and torques (in millinewton-metres) as seven 32-bit floating-point values per sample (F_x, F_y, F_z, T_x, T_y, T_z, and temperature). The electronics was connected to the acquisition computer via its USB interface (12 Mbit/s) appearing as a virtual COM port, and configured at a sample rate of 100 Hz. A custom Python script reads the 28-byte binary packets from the serial stream and logs each sample with a wall-clock timestamp. This sensor was used exclusively in subset 4 (dynamic motions with contact).

Table 1: Summary of the sensor suite. Effective sampling rates account for communication overhead and software processing; native hardware rates may be higher. Selectable parameters show the value used in this work where known; the full set of options is given in the corresponding sensor subsection.

Sensor	**Model / Manufacturer**	**Qty**	**Measured quantity**	**Range**	**Accuracy / Resolution**	**Interface**	**Eff. rate**	**Batches**
Cable tension	Mark-10 M3-5 (Mark-10 Corp., USA)	$\times 4$	Tendon force (1-axis)	0 N to 22.2 N (5 lbf full scale)	$\pm$ 0.3% FS; res. 0.02 N; cal. unc. $\pm$0.007 lbf (k=2)	Serial RS-232 115 200 baud	$\sim$335 Hz	1–4
Shape sensing	FBGS FBG-Scan 904 + 4-core MCF fiber (FBGS Int., Belgium)	1 fiber, 26 DTGs/core	Distributed curvature, bending angle, 3-D shape	50 mm active; 1510 nm to 1590 nm	WL prec. 0.3 pm (1σ); lin. 3 pm; abs. acc. 10 pm; shape res. 1 mm arc	USB; ILLumiSense Shape CORE v1.1	100 Hz	1–4
Base F/T	ATI Mini40 SI-40-2 (ATI Ind. Autom., USA) + NI USB-6210 DAQ	1	6-axis wrench ($F_x, F_y, F_z, T_x, T_y, T_z$) at robot base	$F_{x,y}$: $\pm$40 N; F_z: $\pm$120 N; $T_{x,y,z}$: $\pm$2 N m	16-bit DAQ (250 kS/s); overload F_{xy} $\pm$810 N, F_z $\pm$2400 N	Analog $\pm$10 V via `nidaqmx`; HW clk 1000 Hz, 5-samp. avg.	$\sim$200 Hz	1–4
Motion capture	OptiTrack PrimeX 13 ($\times$5) + PrimeX 13W ($\times$2) (NaturalPoint, USA)	7 cameras; 6 rigid bodies	6-DoF pose (5 disks + 1 object)	Sub-metre workspace; 1.3 MP; 850 nm IR	$\pm$0.15 mm (PrimeX 13); $\pm$0.30 mm (13W); markers $\approx$6 mm dia.	GigE/PoE; Motive 3.4; NatNet 4.4 SDK (unicast UDP)	120 Hz	1–4
IMU	ISM330DHCX (SparkFun / STMicro)	$\times 4$	3-axis angular vel. + 3-axis lin. accel.	Gyro: up to $\pm$4000 dps; Accel: up to $\pm$16 g	16-bit ADC	I^2C / SPI	XXX Hz	3
Actuators	Cybergear QDD (Xiaomi, China)	$\times 4$	Shaft angle (14-bit), velocity, torque	Cont. 4 N m; peak 12 N m; spool r=20 mm	14-bit single-turn absolute encoder	CAN 2.0 1 Mbit/s	XXX Hz	1–4
Contact F/T	Resense HEX 12 (Resense GmbH, Germany) + EVAL 100S-06-1 electronics	1	6-axis wrench ($F_x, F_y, F_z, T_x, T_y, T_z$) at contact point	$F_{x,y,z}$: $\pm$25 N $T_{x,y,z}$: $\pm$125 mN m	1% FS; 3% crosstalk; 24-bit ADC; 300% overload	USB (12 Mbit/s) virtual COM; 28-byte binary packets	1 kHz*	4

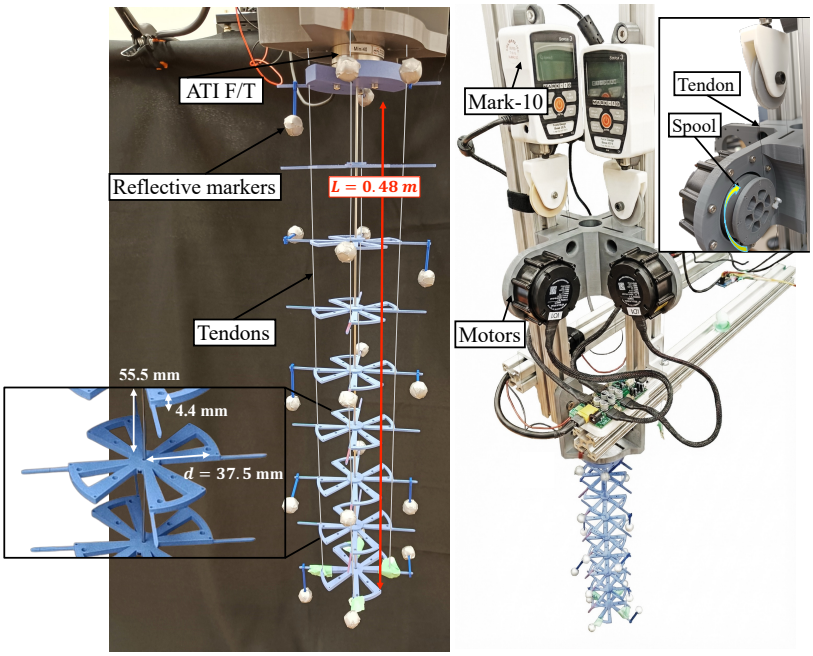


Figure 1: Hardware platform: TDCR with four tendons (two antagonistic pairs), motion capture markers on selected disks, direct-drive motor spools, and inline Mark-10 tension sensing and ATI F/T placed at the base.

2.3 Experimental setup

The robot is mounted on a cantilever platform with its tip pointing downward, *i.e.* in the direction of gravity (see Fig. 1). It is actuated via four tendons arranged in two antagonistic pairs. This layout was chosen as representative of the tendon-driven continuum robots commonly studied in the literature. Each tendon is wound on a spool attached directly to the motor shaft (no gearbox) and routed only through the pulley integrated in the Mark 10 force gauge. Tendon length variations are related to commanded motor rotations through a fixed spool radius ($r_{\text{spool}} = 20$ mm), providing a direct mapping between motor shaft angle and tendon displacement. No additional pulleys or redirections are used, so as to minimise cable friction and eliminate the reflected inertia of intermediate transmission elements. Every other spacer disk is embedded with retroreflective markers and tracked by an OptiTrack motion capture system, providing 6-DOF pose measurements for the tracked disks. In batches 1–3 the workspace is free of obstacles. In batch 4, an extruded aluminium profile is introduced as a contact object. The profile is equipped with two motion capture frames to identify its pose and with a Resense force-torque sensor to measure the interaction force.

Batch	Trajectory type	Parameters	Sensors active	Trials	Duration	Contact

Table 2: Summary of the experiments included in the dataset.

To ensure consistency across all sensor modalities, every recording session follows a fixed preparation protocol.

First, the four tendons are pre-tensioned to approximately 1 N each, with a maximum difference between the two cables of an antagonistic pair <0.2 N. This step minimises the risk of cable slack, a well-known source of modelling error in tendon-driven continuum robots [13].

Second, the recording begins with the robot hanging straight down under gravity. This known reference configuration is used to verify the alignment between the OptiTrack rigid-body frames and the FBGS-reconstructed shape: any offset observed in the straight configuration is attributable to sensor misalignment rather than to robot deformation.

Third, for every non-planar trajectory (*i.e.* circular and Lissajous curves) the robot executes a calibration

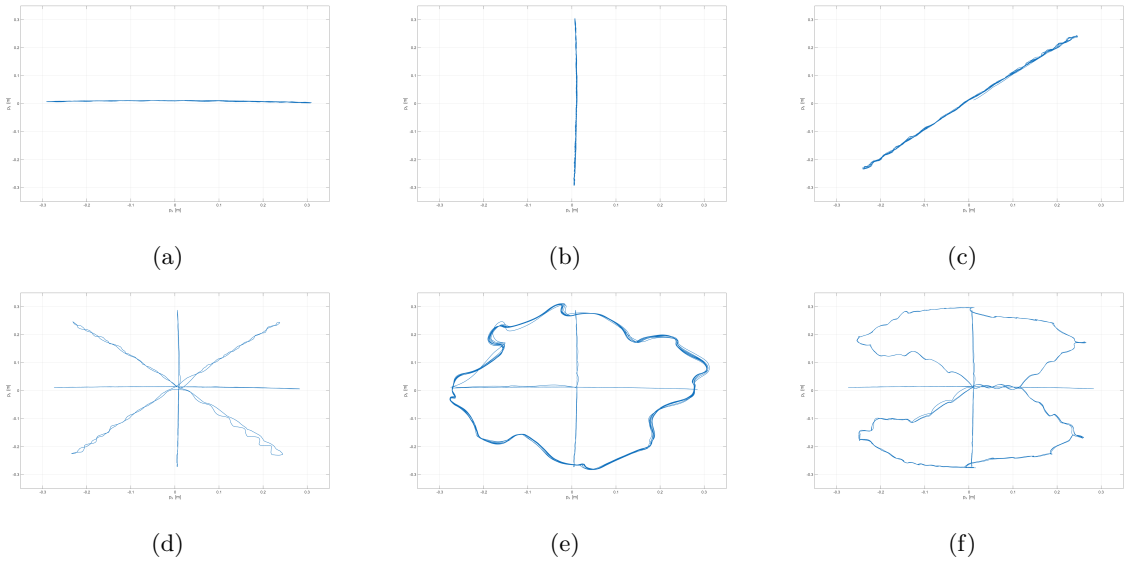


Figure 2: Motion of the robot tip (only xy plane coordinates) for the planar motions on x (a), y (b) and xy (c), the star motion (d), the circular (d) and (f) trajctories.

routine before the target motion. The routine consists of a planar sweep in the x–z plane (positive x followed by negative x), then a planar sweep in the y–z plane (negative y followed by positive y). Because aligning the FBGS fibre to the robot backbone with sub-degree accuracy is difficult in practice, a residual angular offset between the fibre deformation plane and the robot bending plane is expected. The calibration sweeps characterize this offset, allowing it to be corrected in post-processing before the non-planar data is interpreted.

2.4 Data Collection

2.4.1 Acquisition pipeline

For each experiment, a Windows batch script orchestrates the acquisition. The script launches one independent process per sensor, each in its own terminal window. Each sensor process polls its instrument at the native sampling rate (see the sensor specifications in Table 1), buffers all readings in memory, and writes them to a CSV file after the recording is complete. The motor process is launched last, and waits 2 s before sending position commands, providing a sufficient delay to ensure all sensors to be already streaming before the actuators begin to move.

Two sensors require dedicated proprietary software that runs independently from the per-sensor Python acquisition scripts. The FBGS fibre is managed by the ILLumiSense Shape CORE software (v1.1, FBGS), which runs as a background process on the acquisition computer, connects to the FBG-Scan 904 interrogator via USB, performs peak detection on the acquired spectra, and reconstructs curvature, bending angle, and three-dimensional shape in real time. Shape CORE streams its output over a local TCP connection (port 50012). A custom C++ client (`StreamClassTester`) connects to this stream, receives the curvature, angle, and shape arrays at each time step, timestamps each sample using the system high-resolution clock (`std::chrono::high_resolution_clock`), and buffers all data in memory until the recording is complete, at which point it writes a single CSV file. The OptiTrack motion capture system is driven by Motive 3.4 (NaturalPoint, USA) running on a dedicated Windows host PC connected to seven cameras (five PrimeX 13 and two PrimeX 13W) via GigE/PoE. Motive processes and solves the 3-D rigid-body poses at 120 Hz and exposes them through the NatNet 4.4 SDK in unicast UDP mode. A dedicated Python script on the acquisition machine receives each frame over a point-to-point Ethernet link and timestamps it with the wall-clock of the acquisition machine (`time.time()`), ensuring a homogeneous time base across all sensor modalities.

2.4.2 Synchronization

All timestamps in the recorded CSV files are therefore referred to the same operating-system clock, without the need for a hardware trigger. The synchronization accuracy is bounded by the OS thread-scheduling jitter, on the order of 1 ms to 2 ms on the acquisition machine.

2.4.3 Post processing

The raw sensor data is post-processed in MATLAB (`process_data.m`).

For the motion capture data, frames in which one or more markers are occluded are first discarded using the validity flag provided by the tracking software (a value of zero indicates occlusion). The remaining samples are converted from position (x, y, z) and quaternion to SE(3) poses. Because these poses are expressed in the global motion-capture frame, they are re-expressed as transformations relative to the base disk (disk 0). A static rigid-body transformation is then applied to orient the z-axis downward (the gravity direction) and to align the x- and y-axes with the bending-plane convention of the robot.

The FBGS data requires an additional alignment step. The interrogator software reconstructs the fibre shape with the arc-length parameter evolving along its internal x-axis, whereas the physical robot hangs along negative z; a rigid rotation is therefore applied to map the FBGS coordinate frame to the robot body frame. Furthermore, because the fibre cannot be bonded to the backbone with sub-degree angular accuracy, a residual rotational offset exists between the fibre deformation plane and the true robot bending plane. To characterise this offset, a singular-value decomposition is performed on the tip trajectory during the initial planar calibration sweep of each batch: the principal directions of the tip motion identify the bending-plane normal and the primary deflection axis, yielding the correction rotation that aligns the FBGS output to the robot frame.

All signals are then low-pass filtered with a fourth-order Butterworth filter (cutoff frequency 20 Hz) applied in the forward and reverse directions (`filtfilt` in MATLAB) to achieve zero-phase distortion. The filter's normalised cutoff is computed per signal using each sensor's native sampling rate, estimated from the median inter-sample interval. The filtered signals are aligned in time by expressing all timestamps relative to the first actuator timestamp; because every sensor starts recording before the motors, these relative timestamps are initially negative. All signals are then resampled onto a uniform 100 Hz grid spanning $t = 0$ to the end of the actuator recording using linear interpolation (`interp1`). This operation simultaneously discards the sensor preamble (the negative-time samples recorded before the motors started) and brings all modalities to a common time base.

2.4.4 File naming and outputs

Each sensor process writes one CSV file per experiment, with the following naming convention `data{Sensor}.csv`. For example, `dataMotor.csv`, `dataMark10_+x.csv`, `dataATIFT.csv`, `dataOptiTrack.csv`, and `dataFBGS.csv`. Column headers and units are documented in the Data Records section. The post-processed data is saved in a `processed/` subfolder within each experiment directory, containing one file per quantity (`angles.csv`, `cable_tensions.csv`, `base_wrench.csv`, `optitrack_frames.csv`, `fbgs_shapes.csv`), all sampled at 100 Hz with a shared time column. Both the raw and processed datasets are released.

3 Data Records

- Data available on figshare (with DOI) — GitHub for code only
- Define structure dataset
  - Number of folders
  - Number of files, total size
  - Processed and raw data organization
  - Sensors used and how data is organized for each sensor
  - Column-by-column description of every CSV file type

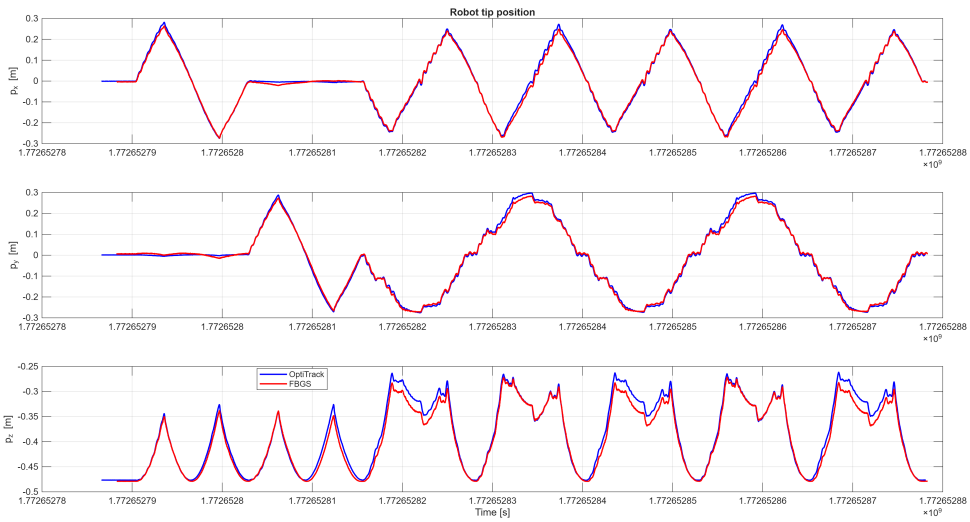


Figure 3: Trajectory of the robot tip as measured by the OptiTrack motion capture system (blue) the FBGS (green) and simulated with the GVS approach fed with tendon tensions (red) using the Lijassous motion.

4 Technical Validation

This section presents the analyses performed to verify the technical quality and internal consistency of the recorded dataset. The dataset contains two families of measurements: kinematic quantities (OptiTrack poses, FBGS strain and reconstructed shape, IMU accelerations, motor encoder angles) and force quantities (Mark10 cable tensions, ATI-FT base wrench, Resense contact force). Within each family, sensors may be directly compared. However, no direct comparison is possible between kinematic and force measurements without invoking the governing physics of the continuum robot. For this reason, a reduced-order dynamics model is employed as a cross-validation tool to verify force–kinematics consistency, as detailed in Section 4.3.

4.1 Sensor noise and calibration

For each experiment, the robots remains static for 3 seconds allowing to obtain an estimate of the static noise for each of the sensors.

Hwever, in this section we provide a static recording of the robot serving as direct measurement of sensors noise and uncertanties.

eg. the mocap frames shift

4.2 Kinematic cross-validation

4.2.1 Comparison of recorded shape with FBGS and Motion Capture

The OptiTrack and FBGS sensors provide independent measurements of the robot shape. To verify their mutual consistency, we compare the tip position recorded by OptiTrack with the tip position reconstructed from FBGS strain data, evaluated at the same arc-length coordinate. Figure 3 shows the overlay of these two signals for the Lissajous trajectory (see Fig. 2f). The overlay show consistency between the two measurements, where deviations could be attributed to the disks drift observed by the static data discussed in Section 4.1. When evaluating the $(x,\ y,\ z)$ position of the robot tip, the RMSE is $(0.0172,\ 0.0140,\ 0.0085)$ m.

4.2.2 Validation of Cable Displacement via Motion Capture

An independent geometric estimate of the cable displacement is derived from the motion-capture data to cross-validate the encoder-based measurements recorded by the actuators. The continuum robot is equipped with four antagonistic cables routed parallel to the elastic backbone and offset by $d = 37.5\,\mathrm{mm}$ from its centreline along the $\pm x$ and $\pm y$ directions in the local cross-sectional frame. Five rigid spacer

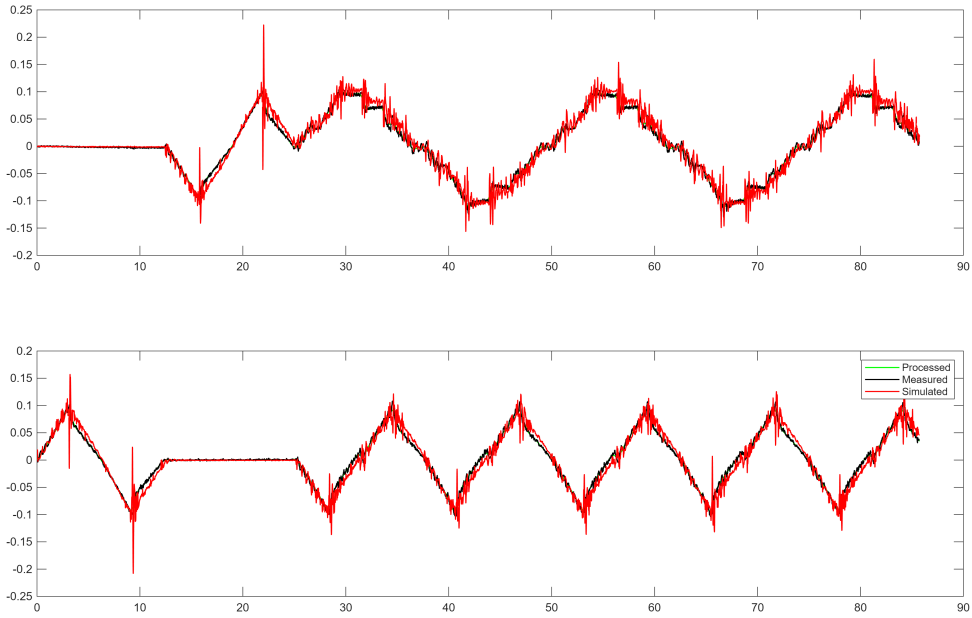
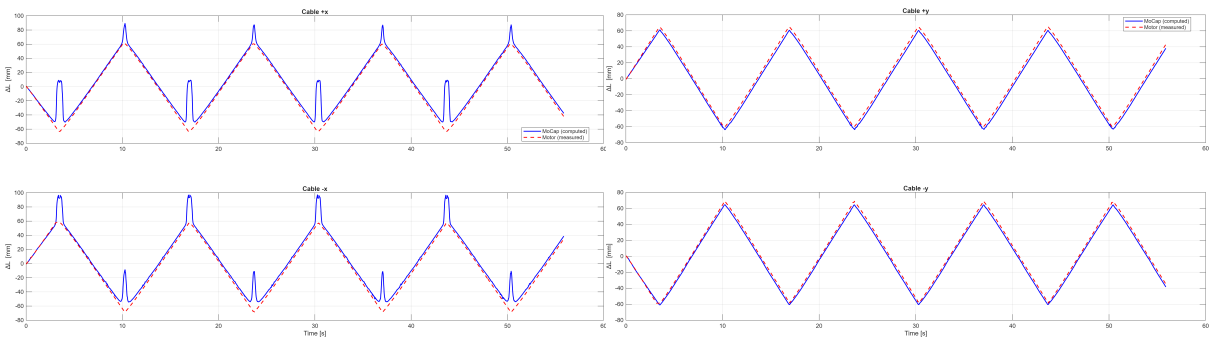


Figure 4: Comparison of the simulated torque at the base (red) with the measured torque by the (unfiltered) ATI FT sensor (green) using the circular motion.



(a) (b)

Figure 5: Comparison of the pulled cable lenght as per measured from the actuator encoders (red) and reconstructed from the OptiTrack recorded kinematics (blue) for the *plane x* (a) and *plane y* (b).

disks, instrumented with reflective markers, are tracked by the optical motion-capture system at 100 Hz and their six-degree-of-freedom poses are reconstructed relative to the robot base frame.

To obtain a dense description of the backbone geometry, the five measured disk frames are interpolated along a normalised arc-length parameter $s \in [0, 1]$ onto 40 uniformly spaced points. Positions are interpolated with a shape-preserving piecewise-cubic Hermite (PCHIP) scheme, while orientations are interpolated by converting each rotation matrix to a unit quaternion, applying PCHIP independently to each quaternion component, and renormalising the result to the unit sphere. Quaternion hemisphere consistency is enforced between successive disks to avoid sign-flip artefacts. At each interpolated frame the world-space position of every cable is computed by applying the local offset vector $\mathbf{d}_c$ through the interpolated rotation matrix. The total length of cable c is then approximated as the piecewise-linear sum of Euclidean distances between consecutive attachment points along the interpolated backbone.

The cable displacement is defined as the change in length relative to the straight (pre-motion) configuration, estimated by averaging the first $N_{\text{ref}} = 10$ frames:

$$\Delta L_c(t) \;=\; L_c(t) - \bar{L}_c^{\,\text{ref}}, \qquad \bar{L}_c^{\,\text{ref}} = \frac{1}{N_{\text{ref}}} \sum_{k=1}^{N_{\text{ref}}} L_c(t_k). \tag{1}$$

An independent estimate is obtained from the motor encoders as $\Delta L_c^{\text{enc}}(t) = \big(\theta_c(t) - \bar{\theta}_c^{\,\text{ref}}\big)\, r_s$, where θ_c is the measured shaft angle and $r_s = 20\,\text{mm}$ is the spool radius. The two estimates are compared directly in the time domain; agreement between them constitutes a geometric consistency check, confirming that the encoder records and the motion-capture poses are mutually consistent and that the assumed cable-routing geometry is correct. Since the scope of this section is to show the consistency of the measured cable lenght with the deformation, we provide a comparison with the motion on the x and y plane. Such comparison is presented in Figure **??**. **Add RMSE**

4.3 Force–kinematics consistency

Cable tensions (measured with Mark10 sensors) and base wrench (measured with ATI-FT sensor) cannot be directly compared against kinematic measurements (OptiTrack, FBGS) without a physical model relating forces to deformations. To verify that these two families of measurements are mutually consistent, we employ a forward dynamics simulation based on the Geometric Variable Strain (GVS) approach [5] as a cross-validation tool.

As the GVS model requires the parameters of the robot to be known, the next section will address how these parameters were roughtly estimated. Then, the following section discusses the comparison of the simulation results with the recorded data.

4.3.1 Rod mechanical parameter characterization

An equivalent bending rigidity EI of the nitinol backbone was estimated in a set of quasi-static holds, where the robot was commanded to three representative bending modes: planar bending in xz, planar bending in yz, and a diagonal bending mode produced by co-activating one tendon from each antagonistic pair. During each hold w, inline force gauges provided the tendon tension vector $\boldsymbol{\tau}_w$, while motion capture reconstructed the robot configuration from markers attached to a subset of spacer disks.

To avoid short settling effects after actuation, the first ~ 200 samples of each recording were removed and the remaining samples were averaged, yielding disk positions $\{\mathbf{p}_i^{\text{meas}}(w)\}_{i=1}^{N_{\text{disks}}}$ in the base frame. For a candidate value of EI, a static Cosserat-rod solver was used to predict the equilibrium shape under the measured tensions $\boldsymbol{\tau}_w$ by enforcing the boundary conditions through a root-finding routine. The simulated disk positions evaluated at the same arc-length locations are denoted by $\mathbf{p}_i^{\text{sim}}(EI, \boldsymbol{\tau}_w)$.

For each bending mode (plane), EI was selected by minimizing the aggregated geometric discrepancy between simulated and measured disk positions over the corresponding set of trials $\mathcal{W}$:

$$\widehat{EI} = \arg\min_{EI} \sum_{w \in \mathcal{W}} \sum_{i=1}^{N_{\text{disks}}} \left\| \mathbf{p}_i^{\text{sim}}(EI, \boldsymbol{\tau}_w) - \mathbf{p}_i^{\text{meas}}(w) \right\|_2^2 . \tag{2}$$

The resulting plane-wise estimates were $\hat{E}I_{xz} = 0.0925$, $\hat{E}I_{yz} = 0.08875$, and $\hat{E}I_{\text{diag}} = 0.0929\ \text{N}\cdot\text{m}^2$; a single value $\hat{E}I = 0.092\ \text{N}\cdot\text{m}^2$ was used in the remainder of this work.

Then, the distributed weight of the rod is calculated from the measured weight of the robot (0.102 Kg) and the geometry of the rod (length 0.48 m and radius 2 mm) producing $\rho = 16910\ \mathbf{kg/m^3}$.

4.3.2 Comparison of the measured forces

Using the GVS of [5] and the characterized robot materials we have a reliable framework to compute the consistency of the measured values. Using the tendon tension as input for our simulation, we can compute the corresponding motion of the robot and the forces it exchanges at its base. Comparison of the computed and measured torque is presented in Fig. 4, were the overall behavior has been captured by the sensor while the simulated spikes are attributed to poorely identified parameters of the robot (*e.g.* the distributed weight and corresponding inertia).

4.4 Contact force validation

To validate the contact force measurements from the Resense FT sensor, we perform a set of static equilibrium checks. A known external force is applied at the robot tip while the robot is held in a static configuration whose shape is recorded by the OptiTrack system. Using the measured shape and the applied force, the corresponding reaction wrench at the base is computed via static equilibrium. This computed wrench is then compared with the direct measurement from the ATI-FT sensor at the base.

4.5 Temporal synchronization

All sensors are acquired as concurrent threads in a single Python process on a single machine. Each thread timestamps its readings using the operating system wall-clock (`time.time()` in Python), ensuring that all sensors share the same clock source without requiring hardware synchronization. The resulting synchronization error is bounded by the operating system thread scheduling jitter, which is on the order of 1 ms to 2 ms on the acquisition machine, well below the signal periods of interest (the highest dynamic content in the dataset is below 20 Hz).

5 Usage Notes

Data has been processed using MATLAB R2025b. During the postprocessing, the MATLAB functions: *butter*, *filtfilt* and *interp1* were used. However, no restriction is applied on the used software as long as reading a processing *.csv* files is possible.

6 Data Availability

Many people in Nature Scientific Data use figshare for which I have created an account.

7 Code Availability

Statement about releasing the code for the data acquisition and the postprocessing & technical validation.

8 Author Contributions

9 Competing Interests

The authors declare no competing interests.

10 Funding

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